

CCS 239 OPTIMIZATION THEORY *and Application*

An Interactive Optimization and Root-Finding Tool

I. Areas of Focus:

Choose one from the following methods to implement:

1. Bisection Method for Root-finding
2. Bisection Method for Optimization
3. Fixed Point Iteration Method
4. Golden Section Search Method
5. Fibonacci Search Method
6. Newton-Raphson Method for Root-finding
7. Newton-Raphson Method for Optimization
8. Secant Method

II. Task:

Implementation of the method chosen by developing an application/program you are most comfortable with, for example, Python via Streamlit, Tkinter, etc., MS Excel, website, etc.

Objective:

Create a user interface that allows the user to input polynomial functions of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

If you could allow the user to input other mathematical expressions such as $\sin x$, $\cos x$, e^x , the better.

Furthermore, the application act as an interactive tool where users can:

1. Input Fields:

- Input coefficients for the polynomial or input the polynomial itself.
- Specify the left and right endpoints for the interval (if applicable).
- Define the tolerance level for stopping/ending the iterations.
- Any other necessary values to ensure the algorithm functions properly (e.g., initial guesses for some methods).

2. Results Generation:

Once the necessary inputs are provided, it should:

- Compute and display the number of iterations for the chosen method within the tolerance indicated.
- Compute and display the approximate root or minimizer.
- IF POSSIBLE, display a table with values from each iteration (e.g., intervals, function values, errors). Just like what we did in MS Excel.

3. Graphical Output:

- The application should graphically display the function over the specified/chosen interval or in an interval where the root/minimizer is located.

III. Groupings, Customization and Layout:

You are free to have your own grouping, 3-4 members per group, and no duplication of chosen method. Be mindful of your groupings and see to it that each group has a balanced composition. You are free to design and style the layout in a way that it is user-friendly, visually appealing, and functional.

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[PROJECT TITLE (All uppercase letters)]

[Team Member (Last Name, First Name Middle Initial)]
[Team Member (Last Name, First Name Middle Initial)]
[Team Member (Last Name, First Name Middle Initial)]

Project Overview:

Provide a brief description of the project. Include: (a) a summary of the selected algorithm(s), (b) how the algorithm works conceptually (e.g., finding roots, minimizing functions), and (c) How the program implementation automates the process for the user.

Objectives/Features:

Clearly state the main objectives of the project.
Enumerate specific goals or outcomes expected from the program.

Criteria for Scoring/Grading:

Documentation and Presentation (10%)	• Clear and well-organized documentation that explains the functionality of the program. • Proper packaging of the documentation, including the sections listed above.
Functionality and Accuracy (70%)	• The program must work correctly for the chosen algorithm(s). • It should accurately compute the iterations and output the correct root or minimizer. • The calculations must reflect the user-defined inputs (coefficients, intervals, tolerance, etc.).
Design and Layout of the Excel Sheet (20%)	• The user interface should be well-organized and easy to navigate. • Ensure that the layout is visually appealing, with clear instructions for users. • The graphical output (e.g., function plot) must effectively illustrate the process of finding the root or minimizer.

Additional Note to Students:

Flexibility in Design:
You are encouraged to customize the interface of the application/program according to your preferred style. Creativity in design is welcome, as long as it does not compromise functionality.

Use of Polynomial and Other Mathematical Functions:
Ensure the input fields are flexible enough to handle polynomials and other mathematical expressions.

Interactive User Interface:
The goal is for the user to interact with the tool by simply providing the necessary inputs, and the application/program handles the rest, from results generation to graph(s).